

GeoFDI: Fault and Slip Detection as Invariance Testing, with Finite-Sample Guarantees for Legged and Wheeled-Legged Robots

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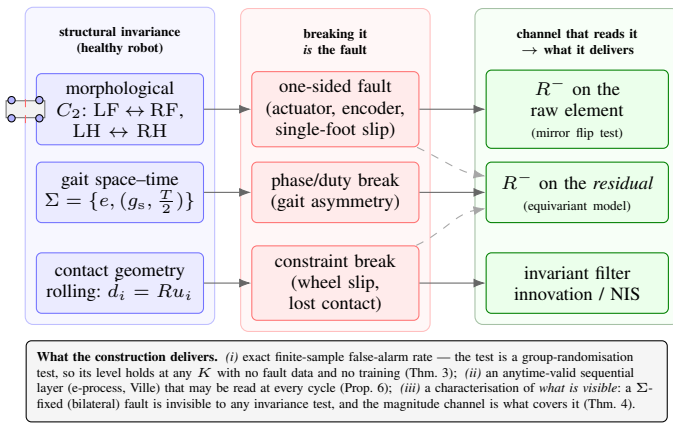


Fig. 1. **Fault and slip detection as invariance testing.** A healthy legged or wheeled-legged robot satisfies three structural invariances (left): the morphological mirror symmetry of the body, the space-time symmetry of the commanded gait, and the geometry of contact (a stance foot is fixed; a rolling wheel’s contact point moves at the measured rolling rate). Every fault or slip we target *is* a break of one of them (middle), which turns detection into a test of a symmetry hypothesis rather than a comparison against a learned or thresholded nominal. Three channels read the break (right); dashed links mark the couplings that make them complementary rather than redundant.

Abstract—[abstract placeholder]

I. INTRODUCTION

A legged robot’s usefulness outside a laboratory rests on two things it cannot observe directly: whether its actuators still deliver what the controller asks, and whether its feet are where the estimator believes they are. Both degrade silently. An actuator loses torque gradually; a foot slips for a fraction of a step on a wet tile; an encoder drifts by a few milliradians. None of these announce themselves, and each corrupts the state estimate and the controller that depends on it.

The established answers fall into two families, and both leave the same hole. *Model-based* residual methods run a dynamics model — a generalised-momentum observer is the canonical choice [1], [2] — and threshold the residual, usually against a χ^2 quantile. *Data-driven* methods learn what nominal

looks like, from an autoencoder or a recurrent classifier, and threshold a reconstruction error or a predicted fault parameter. The hole is the false-alarm rate. A χ^2 threshold is exact only if the residual is i.i.d. Gaussian, and a legged robot’s residual is neither: it is gait-periodic, serially correlated, and non-stationary across a recording. We measure the consequence directly. On our simulation benchmark the textbook momentum-observer χ^2 detector runs at a nominal alarm rate of **0.15** against a nominal 0.05, and re-calibrating its threshold on the robot’s own nominal data does not repair it — on a *held-out* stretch of the same nominal recording the rate is **0.50**, because the residual’s statistics have already moved (§VI). On an external, third-party fault dataset a Mahalanobis detector fires on **52 %** of healthy segments (0.522 false alarms per healthy gap) where our test fires on **4.9 %**, i.e. at its nominal level (§VII). A detector whose false-alarm rate is unknown cannot be given authority over a controller, which is why slip handling in practice is still a hand-tuned threshold on foot speed [3].

a) Gap: Three literatures each hold one piece and none holds the combination. Morphological-symmetry work in robotics [4] uses a robot’s discrete symmetry group to constrain models and policies, but treats symmetry as an architectural prior, not as a hypothesis to be tested. Fault detection and contact estimation [5], [6] build ever better nominal models and contact classifiers, but calibrate their decisions empirically. Invariance testing in statistics [7], [8] gives exactly the guarantee the robotics side lacks — a group-randomisation test whose level is exact in finite samples, and an anytime-valid sequential form — but has not been connected to a physical invariance a robot actually possesses. The missing step is to notice that a healthy legged robot *already satisfies* a group invariance precisely enough to serve as the null hypothesis, and that the faults we care about are exactly the things that break it.

b) Thesis: We propose to detect faults and slip by testing the structural invariances of the robot itself (Fig. 1). The morphological mirror symmetry of the body, the space-time symmetry of the commanded gait, and the geometry of

contact are properties a healthy machine must satisfy; a one-sided actuator fault, a single-foot slip, or a broken rolling constraint are, by construction, violations of them. Framing detection this way replaces a learned or thresholded nominal with a hypothesis that needs no fault data, no training, and — in its model-free form — no dynamics model, and it inherits an exact finite-sample false-alarm rate from the group-randomisation construction.

c) Contributions:

- 1) **Invariance as the null, with an exact level.** We formulate the healthy-robot hypothesis H_0 as distributional invariance of a phase-registered data element under the gait symmetry group, and test it with a group-randomisation (flip) test whose false-alarm rate is *exact at any sample size*, plus an anytime-valid sequential layer built from e-processes. We also give the recalibrated null H'_0 for the realistic case of a robot that is stably but measurably asymmetric, and a power lower bound with an explicit sample-size rule (§IV; Thm. 3, Prop. 5, Prop. 6).
- 2) **A two-layer characterisation of what is visible, and the architecture it forces.** We prove a dichotomy: a fault fixed by the symmetry group (a bilaterally symmetric degradation) is invisible to *every* invariance test, whatever its magnitude, provided the faulted loop keeps a unique symmetric attractor; anything that breaks the law is detectable by a consistent statistic. The mean-level layer gives the power of the specific paired statistic, and the two layers do not coincide — a zero-mean law change is invisible to the paired statistic and visible to an energy-distance one. This is why the system needs three channels rather than one (§IV; Thm. 4).
- 3) **Equivariant nominal models, and why equivariance is not optional.** When a learned dynamics model is used to form a residual, we show the residual inherits H_0 only if the model is equivariant, quantify the contamination a non-equivariant model injects, and construct an equivariant Deep Lagrangian Network by mirror weight sharing. The effect is not marginal: a plain learned model drives the residual test’s false-alarm rate to 1.00 while its equivariant twin stays at 0.00 (§IV, §VI).
- 4) **Rolling contact preserves the invariant filter, and a five-layer evidence stack.** We prove that a rolling wheel’s moving contact keeps the contact-aided invariant filter group-affine — the invariant error dynamics is unchanged, only the mean and the process noise move — which admits wheeled-legged robots to the same estimator and the same gating front-end as point-foot robots. The claims are evaluated at five levels of increasing externality, including *two* of our own hardware platforms (§VIII).

d) What the evaluation shows: On our own Unitree Go2 corpus — eleven sessions, a machine over a year in service, three sites, two dates two months apart — the naive invariance test rejects on **11/11** sessions, i.e. it sees the machine’s real,

stable left–right asymmetry, and the recalibrated level ν_0 reproduces across the two-month gap to within a factor **1.18**, which is smaller than the session-to-session spread within a single day. On the wheeled-legged M1 the rolling-contact filter recovers **0.99–1.00** of the driven path length where a fixed-foot filter recovers **0.03–0.04**. Used as an estimator front-end, the calibrated per-event gate discards **1.9 %** of nominal contact measurements where the standard fixed foot-speed threshold discards **40.6 %** of them on the same real outdoor data. And on the external benchmark the test holds its level at **0.049** against a nominal 0.05.

II. RELATED WORK AND POSITION

a) Model-based fault and collision detection: The generalised-momentum observer [1] and the family of residual generators surveyed in [2] are the reference point: they need no fault data, they isolate to a joint, and they are fast. Their decision rule is a threshold, usually a χ^2 quantile of a whitened residual, which is exact under an i.i.d. Gaussian assumption that a periodic legged system violates. We reproduce the method faithfully and measure the gap (§VI); the point of comparison in this paper is never raw sensitivity — the classical detector is often *more* sensitive — but whether the stated false-alarm rate is the delivered one.

b) Contact estimation and slip handling: Probabilistic contact estimation from foot-mounted inertial sensing [5] and learned contact classifiers [6] improve the input a state estimator receives; controllers for slippery ground [3] act on detected slip. These decide contact or slip; they do not certify how often they are wrong on healthy data. Our gate targets exactly that quantity, and we compare against the standard fixed foot-speed threshold on real outdoor data (§VIII).

c) Symmetry in robotics: Morphological symmetry groups have been used to constrain learned dynamics and control [4], and adaptation methods recover from damage without diagnosing it [9]. Both treat symmetry (or its loss) as structure to exploit; neither turns it into a hypothesis with a level.

d) Invariance testing: Group-randomisation tests give exact finite-sample validity for invariance nulls [7], and recent work provides anytime-valid *e*-value forms for group invariance [8], [10]. Our contribution on this side is not a new test but the identification of a physical invariance — of a robot, under its own morphology and gait — that these tests can be applied to, together with what the resulting detector can and cannot see.

Table I is the summary. Read row-wise it is not a claim that the invariance test dominates: the classical observer is faster and, at the smallest friction magnitudes, more sensitive; the GRU isolates to a joint and detects everything it was trained on. Read column-wise, the three columns that no baseline fills — a false-alarm rate that is exact rather than empirical, immunity to symmetric nuisances, and a sequential guarantee that survives continuous monitoring — are the ones a robot needs before a detector is allowed to interrupt a controller.

TABLE I
WHAT EACH DETECTOR REQUIRES AND WHAT IT DELIVERS. EVERY CELL IS MEASURED IN THIS PAPER; THE EXPERIMENT THAT PRODUCED IT IS NAMED. “FAR HELD” MEANS THE MEASURED NOMINAL ALARM RATE MATCHES THE STATED LEVEL.

detector	fault data	training	dynamics model	finite-sample FAR held	symmetric-nuisance immune	sequential guarantee
GeoFDI R^- (raw)	no	no	no	yes (0.00, e21; 0.049 ext., e03)	yes (0.00, e07/e21)	yes (Prop. 6)
GeoFDI R^- (residual)	no	no	yes	yes (0.00, M0.3)	yes (inherits H_0)	yes
momentum observer + χ^2	no	no	yes	no (0.15 fixed / 0.50 recal., e21)	no (0.90–0.95, e21)	no
Mahalanobis	no	nominal only	no	no (0.52 per healthy gap, e03)	no (0.50, e07)	no
autoencoder	no	nominal only	no	not stated	no (0.50, e07)	no
GRU classifier/regressor	yes	yes	no	empirical only (0.00–0.012, e03)	no (0.65, e07)	no

III. PRELIMINARIES

a) *Symmetry group and representation:* Let $G = \mathbb{C}_2 = \{e, g_s\}$ be the sagittal (left–right) reflection of the robot, acting on leg labels by $g_s : (\text{LFRF})(\text{LHRH})$. Front–hind symmetry is *not* assumed: mass distribution and leg geometry differ between the pairs, so the richer $\mathbb{K}_4$ of idealised quadrupeds is unavailable. On the telemetry channel space the reflection acts by a signed permutation $\rho_W(g_s)$: a channel maps to its mirror partner with a sign fixed by whether the quantity is polar or axial about the sagittal plane. Joint angles about the roll axis flip; pitch-axis joints do not; a foot position or velocity in the body frame is polar, so $(x, y, z) \mapsto (+x, -y, +z)$ of the partner leg; a foot force is a magnitude; the IMU accelerometer is polar and its gyroscope axial. All signs used here were verified on hardware telemetry rather than assumed from a datasheet.

b) *Gait group and data element:* A commanded gait adds a temporal symmetry. For a trot the relevant group is the *space–time* group $\Sigma = \{e, \sigma_*\}$ with $\sigma_* = (g_s, \frac{1}{2})$: mirroring the robot and shifting by half a gait period maps the healthy motion to itself. Telemetry is phase-registered into a *data element* $Z_k \in \mathbb{R}^{d \times N}$: one gait cycle of d mirror-paired channels resampled on N phase points. For a robot without a gait phase — a wheeled-legged platform in rolling mode — the temporal factor is absent, $\Sigma = G$, and the element is a fixed-duration block. The group acts on elements by $(\rho(\sigma)Z)(\theta) = \rho_W(g)Z(\theta + \Delta\theta)$, an orthogonal involution, so the element space splits into the isotypic components

$$\Pi^\pm = \frac{1}{2}(I \pm \rho), \quad \mathcal{W} = \mathcal{W}^+ \oplus \mathcal{W}^-, \quad (1)$$

the mirror-symmetric and mirror-antisymmetric parts. $\mathcal{W}^-$ is what an invariance test can see; $\mathcal{W}^+$ is what it is structurally blind to.

c) *The null hypotheses:*

Definition 1 (H_0 , healthy invariance). $H_0 : \rho(\sigma)_\# P = P$ for every $\sigma \in \Sigma$, i.e. the law of the data element is invariant under the gait symmetry.

H_0 is what an ideal machine satisfies. A real machine, in service for a year and carrying a payload, is stably but measurably asymmetric, and H_0 is then false for a healthy robot — as we confirm on every one of our eleven Go2 sessions. The deployable null is therefore the recalibrated one:

Definition 2 (H'_0 , unchanged asymmetry). Let $\nu(P) \geq 0$ be an asymmetry functional (the energy distance between $\{Z_k\}$ and $\{\rho Z_k\}$ on standardised channels), which vanishes iff the law is mirror-invariant. $H'_0 : \nu(P_{\text{mon}}) = \nu(P_{\text{cal}})$ — the asymmetry level of the monitored window equals the level calibrated on nominal data.

H_0 is a special case ($\nu_0 = 0$). The standing assumptions behind both — equivariant dynamics, noise and command; a unique symmetric attractor; and exchangeability of elements — are stated with their ε -approximate variants and their audits in Appendix A.

IV. THEORY

This section states the results and the intuition; every proof is in Appendix A, which also carries the map from the numbering here to the numbering of the technical notes.

A. *An exact test, at any sample size*

Theorem 3 (Exactness of the flip test). *Let $Z_1, \dots, Z_K$ be data elements that are exchangeable under H_0 , and let T be any statistic. Draw $M - 1$ i.i.d. uniform group elements $\sigma_m \in \Sigma$ (equivalently, i.i.d. signs on the antisymmetric component) and let $p = \frac{1}{M}(1 + \#\{m : T(\rho(\sigma_m)Z) \geq T(Z)\})$. Then under H_0 , $\mathbb{P}(p \leq \alpha) \leq \alpha$ for every K and every M , with equality when αM is an integer.*

The mechanism is that under H_0 the observed configuration and its group images are equally likely, so the rank of the observed statistic among its own orbit is uniform. Nothing is estimated, nothing is fitted, and no asymptotics are invoked: the level holds at $K = 10$ as it does at $K = 1000$. The price is that the test only sees the group — which is exactly what the next theorem characterises.

B. *What is visible: a two-layer characterisation*

Theorem 4 (Two-layer detectability). (Layer I, law level.) *If the fault pattern is fixed by Σ (a bilaterally symmetric degradation) and the faulted closed loop retains a unique symmetric attractor, then $\rho_\# P_\varphi = P_\varphi$ and every level- α invariance test has power $\leq \alpha$, whatever the magnitude. Conversely, if $\rho_\# P_\varphi \neq P_\varphi$, a statistic consistent for the two-sample problem between $\{Z_k\}$ and $\{\rho Z_k\}$ — the energy-distance class — has power $\rightarrow 1$. (Layer II, mean level.) For the paired-energy statistic the noncentrality after K elements*

is $K\|\Pi^-\mu(\varphi)\|_\Sigma^2$, and the test is consistent iff $\Pi^-\mu(\varphi) \neq 0$ — a strict subset of the laws Layer I covers.

Three consequences drive the system design. First, blindness to bilaterally symmetric faults is structural, not a weakness of a particular statistic, so a second, magnitude channel reading $\mathcal{W}^+$ is *necessary*, and it cannot inherit the same nuisance immunity. Second, the two layers differ exactly on zero-mean law changes: a one-sided variance inflation has $\Pi^-\mu = 0$ but breaks the law, so the paired statistic is blind while the energy-distance one is consistent — deployment runs both. Third, the blindness is robust in amplitude: sweeping a bilateral fault down to 30 % of nominal torque left the test at its level and the realised gait’s chirality at its nominal floor, so the escape route is a genuine symmetry-breaking bifurcation of the gait, not a large enough fault.

Proposition 5 (Power and sample size). *Under a Gaussian model for the differenced scores, the power of the whitened quadratic flip statistic at level α is at least $1 - \exp(-(d + \lambda - c_\alpha)_+^2 / (4(d + 2\lambda)))$ with $\lambda = K\|\Pi^-\mu\|_\Sigma^2$, which inverts to an explicit sample-size rule $K \geq \lambda^*(d, \alpha, \beta) / \|\Pi^-\mu\|_\Sigma^2$.*

C. Reading it every cycle

Proposition 6 (Anytime-valid sequential layer). *Let $p_1, p_2, \dots$ be the window p -values of Thm. 3 and $e_t = \frac{1}{2}p_t^{-1/2}$. Then $E_t = \prod_{s \leq t} e_s$ is a nonnegative supermartingale with $E_0 = 1$, and by Ville’s inequality the rule $E_t \geq 1/\alpha$ has false-alarm probability at most α over an unbounded horizon.*

This is what makes the test deployable: it may be inspected after every cycle, and stopping when it looks interesting does not invalidate it.

D. Residual channels and equivariance

Running the same test on the residual of a nominal dynamics model, $r = y - \hat{f}(x, u)$, both sharpens it and endangers it.

Proposition 7 (Residual inheritance and its contamination). *If the nominal model $\hat{f}$ is Σ -equivariant, the residual element inherits H_0 exactly, and the flip test on it retains its level while gaining power against faults that act through the dynamics. If $\hat{f}$ is not equivariant, its equivariance defect enters the residual as a fixed asymmetric offset and inflates the level.*

The effect is large enough to matter in practice, not merely in principle: on the welded-leg world a plain learned model drives the residual test’s size to 1.00 while its equivariant twin sits at 0.00 (§VI). Mirror weight sharing is what makes a Deep Lagrangian Network equivariant by construction, and the equivariance defect is then exactly zero rather than small.

E. Rolling contact keeps the invariant filter

Proposition 8 (Rolling contact is group-affine). *For a wheel in contact with rolling constraint $\dot{d}_i = Ru_i$, u_i a measured input, the contact-aided invariant filter’s propagation remains group-affine, and the right-invariant error dynamics of the contact block is unchanged: $A[d_i, R] = 0$, exactly as for a world-fixed foot. Rolling alters only the mean propagation $d_i \mapsto$*

Algorithm 1 GeoFDI tester (one session)

- 1: **segment**: cut telemetry into steady, straight, single-condition runs
 - 2: **register**: estimate gait phase; resample each cycle onto N points $\rightarrow$ elements Z_k (rolling mode: fixed-duration blocks)
 - 3: **calibrate**: on the first K_{cal} elements of *this run*, estimate the asymmetry level ν_0 and the channel scales
 - 4: **test**: per window of w elements, flip test (Thm. 3) with statistics {paired-energy, energy-distance} $\rightarrow p_t$
 - 5: **accumulate**: $e_t = \frac{1}{2}p_t^{-1/2}$, $E_t = \prod_{s \leq t} e_s$; alarm if $E_t \geq 1/\alpha$ (Prop. 6)
 - 6: **recalibrate**: on hardware, replace H_0 by H'_0 — test $\nu_{\text{mon}} = \nu_{\text{cal}}$
 - 7: **isolate**: rank mirror *pairs* by Π^- energy; obtain the *side* from the signed mean
 - 8: **gate** (optional): per-stance-event e -values $\rightarrow$ per-leg weights $\pi_i \rightarrow$ invariant-filter update
-

$d_i + Ru_i\Delta t$ and the contact process noise, which becomes anisotropic in the wheel frame.

The plain-EKF Jacobian for a moving world point is $-R(u_i)_\times$, and carrying it into an invariant filter is a natural and wrong step that makes the error dynamics state-dependent. The correct statement admits wheeled-legged robots to the same estimator, and hence the same gating front-end, as point-foot robots. The signature of a fault in such a filter is the observable projection of its direction transported by the adjoint, which is why a bilaterally symmetric slip is invisible to the invariance channel but visible to the estimator channel — the property that turns the pair into a slip *classifier* (§V).

V. METHOD

A. Three channels

The detector is three readings of the same telemetry, forced by Thm. 4 rather than assembled by taste. R^- tests invariance on the raw element and sees exactly the antisymmetric component $\mathcal{W}^-$: one-sided faults and one-sided slip, with an exact level and no model. R^+ reads the symmetric component $\mathcal{W}^+$ of a model residual and covers what R^- is structurally blind to — bilaterally symmetric degradation — at the cost of needing a dynamics model and of inheriting no nuisance immunity. The estimator channel reads the innovation of the contact-aided invariant filter, which responds to constraint violations (slip, lost contact) whether or not they are symmetric. Running R^- on the *residual* rather than the raw element is a fourth combination and is what makes friction-type faults visible to the invariance test at all (§VI).

B. The tester

Algorithm 1 is the whole procedure. Its correctness rests on nine conditions, listed beside it; each is either a theorem hypothesis or an empirical calibration, and each is tagged with where it is established.

TABLE II
THE NINE CONDITIONS ALGORITHM 1 DEPENDS ON, AND WHERE EACH IS
ESTABLISHED.

#	condition	established by
C1	the plant, command and noise are Σ -equivariant	assumptions A1–A4, audited
C2	the healthy loop has a <i>unique</i> symmetric attractor	A5; falsified by a chiral gait at low gain
C3	elements are exchangeable across the index	Assumption E; block length $\geq 2 \times$ the nuisance correlation time
C4	the phase registration is half-period equivariant	$\pm 10\%$ clock error costs power, not level
C5	the segment is a single homogeneous condition	per-run calibration (§V-C)
C6	on hardware, use H'_0 : the level ν_0 , not $\nu = 0$	Def. 2; 11/11 sessions reject H_0
C7	a residual channel needs an <i>equivariant</i> model	Prop. 7; plain model $\rightarrow$ size 1.00
C8	bilaterally symmetric faults need the R^+ channel	Thm. 4 (Layer I)
C9	side attribution needs a signed statistic	energy statistics rank mirror <i>pairs</i> only

C9 deserves emphasis because it is easy to get wrong and we did get it wrong once. For a mirror pair (c, c') the antisymmetric component satisfies $(\Pi^- Z)[c'] = -s(\Pi^- Z)[c]$, so the two partners carry *identical* Π^- energy: no energy-type statistic can say which side is at fault, only which pair. Standardising each channel by its own scale before summing breaks that identity numerically and manufactures a spurious side ranking. The side is recoverable only from the *sign* of the mean difference, which is what Algorithm 1 line 7 uses.

C. Calibrate per traverse, not per session

The single most consequential deployment rule we found is not in the theory but in the hardware data. On both of our platforms the recalibrated null H'_0 holds *within* a continuous traverse and fails *across* traverses: on the Go2 corpus the sequential monitor alarms on 8 of 11 sessions when the element pools several straight runs, but on only 2 of 11 when it is built from a single run, and a shape classifier labels 8 of 11 ν -trajectories as boundary-jumps and none as drift. The same signature appears on the wheeled M1. The mechanism is that the healthy asymmetry level depends on heading, ground and payload placement, so concatenating traverses violates the premise that calibration and monitoring windows come from one regime. The rule is therefore: *calibrate ν_0 per homogeneous segment*. A segment is not automatically homogeneous merely because it is one continuous run — §VIII reports a session where one spatial half is internally clean and the other is not.

D. Equivariant nominal model and π_i gating

The learned nominal model is a Deep Lagrangian Network made equivariant by mirror weight sharing: one template per leg pair, applied to the mirrored coordinates of the partner leg,

so the equivariance defect is exactly zero rather than small. For the estimation front-end, each stance event (a touch-down and the contact interval it opens) is scored by the pre-update per-foot innovation, converted to a conformal p -value against a library of nominal events, and turned into an e -value; the per-leg e -values become weights π_i that either drop (hard gate, $e \geq 1/\alpha$) or down-weight (soft gate, covariance scaled by $1/w_i$) that foot’s measurement. Because the gate is calibrated rather than thresholded, its per-event false-alarm rate is $\approx \alpha$ by construction — which is the property the standard fixed foot-speed threshold lacks (§VIII).

VI. SIMULATION STUDY

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VII. EXTERNAL BENCHMARKS

[placeholder]

VIII. HARDWARE EVALUATION

[placeholder]

IX. LIMITATIONS

[placeholder]

X. CONCLUSION

[placeholder]

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APPENDIX A PROOFS

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